

Igor Tsarev

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Education

University of California, San Diego

B.S. in Computer Science

La Jolla, CA

Expected June 2030

Hackathons & Awards

Leaders of Digital Transformation Hackathon – 1st Place, X5 Group Track

2025

Annual hackathon hosted by the Moscow City Government

- Ranked 1st of 61 teams with F1 = 0.937 at 82.5 ms latency.
- Built a custom tokenizer and training pipeline and benchmarked base language models for X5 Group, Russia's largest grocery retailer.

AIJC AI Challenge – Prize Winner, T-Bank Track

2024

Team Lead | International AI competition

- Led a team building *SpeechApp*, a speech-practice app that detects rhotacism (difficulty pronouncing the “r” sound) in recorded speech and offers guided exercise courses with progress tracking.
- Trained a 1D-CNN classifier on MFCC, mel-spectrogram, chroma, zero-crossing-rate, and RMS features, using noise, pitch-shift, and time-stretch augmentation; built a FastAPI backend, a Next.js PWA, and a Telegram bot client.

Central University Bootcamp – 1st Place

2024

Team Lead

- Led a team building a ChatGPT-powered Telegram bot that runs interactive games in group chats.

Projects

Checko – AI Receipt Processing

[GitHub](#)

Python, FastAPI, React, TypeScript, Tesseract OCR, LLMs, Sentence-Transformers, Docker, Nginx

- Built a Telegram Mini App that turns receipt photos and PDFs in 7 formats into structured expense records using Tesseract OCR and LLM extraction into schema-validated JSON.
- Implemented semantic search over each user's receipts using sentence-transformer embeddings and cosine similarity.
- Developed the FastAPI backend with background processing and the React/TypeScript frontend; packaged the app with Docker Compose, Nginx, and Certbot (HTTPS) for deployment.

Face Recognition Logger

[GitHub](#)

Python, OpenCV, dlib, PyQt5, PostgreSQL

- Built a desktop app that detects faces in a live camera feed with an OpenCV DNN (SSD) detector and identifies them against a gallery of known people using dlib face encodings, parallelizing matching with a thread pool.
- Logged each detection with a timestamp and snapshot to PostgreSQL and added daily and monthly report views to the PyQt5 interface.

Courses & Certifications

- **Relevant coursework (self-study):** Stanford CS229 (Machine Learning), Stanford CS230 (Deep Learning), Harvard CS50AI
- **SAT Math:** 800/800
- **Codeforces:** rating 1546 (Specialist)
- **Machine Learning Specialization**, Yandex 2024
- **Python Programming**, Yandex Lyceum (2-year program) 2022 – 2024

Skills

Programming	Python, C++, Java, SQL
ML & Data	PyTorch, NumPy, Pandas, OpenCV
Backend & Tools	FastAPI, PostgreSQL, Docker, Git, Linux
Languages	English (fluent), Russian (native), Ukrainian (native)